EXTREMALITY OF TRUNCATED BERGMAN FANS IN NEF CONES OF PERMUTOHEDRAL VARIETIES

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ABSTRACT. Truncated Bergman fans $\Delta_M[r_1, r_2]$ of a matroid M are Minkowski weights on the permutohedral fan whose classes are nef on the permutohedral variety. They were introduced by Huh, who observed that they often, but not always, span extremal rays of the nef cone. We study when this happens. For one-dimensional truncations we give a complete answer: $\Delta_M[j, j]$ is extremal if and only if $j = 1$, or $j = \operatorname{rk} M - 1$ and the lattice of flats of M is modular. In higher dimensions we show that extremality of $\Delta_M[r_1, r_2]$ does not depend on $r_2 > r_1$, and that it is equivalent to the uniqueness, up to scaling, of the solution of an explicit linear system on the flats of ranks 1, r_1 and $r_1 + 1$. As consequences, $\Delta_M[1, r_2]$ is always extremal, and $\Delta_M[r_1, r_2]$ is extremal whenever every restriction of M to a flat of rank $r_1 + 1$ has a modular lattice of flats. In the other direction we give a combinatorial sufficient condition for non-extremality and use it to show that the graphic matroids $M(K_{2,m})$, $m \geq 3$, give an infinite family of non-extremal truncated Bergman fans. An exhaustive computation over all loopless matroids on at most seven elements shows that the smallest example is $M(K_{2,3})$, which we identify with an example in Huh's thesis, and that every non-extremal example in this range is a small modification of $M(K_{2,3})$.

1. INTRODUCTION

Let $E = \{0, 1, \ldots, n\}$ and let $X_E = X_{A_n}$ be the permutohedral variety, the smooth projective toric variety of the permutohedral fan Σ_E, which plays a central role in the Hodge theory of matroids [1]. Nef classes on X_E of arbitrary codimension are, by the work of Fulton and Sturmfels [6], the same as nonnegative Minkowski weights on Σ_E, and the cone they form is rational polyhedral. Describing its extremal rays is believed to be difficult even for small n (see [20]). For divisors it is the cone of submodular functions on the Boolean lattice of E modulo modular functions [17], whose extremal rays are known only for very small $|E|$; for instance there are 37 of them when $|E| = 4$ [18].

Matroids provide natural extremal rays. Huh showed that the Bergman fan Δ_M of a loopless matroid M on E spans an extremal ray of the nef cone of the appropriate dimension [12, Thm. 5.7], [11]; the divisorial analogue is Nguyen's earlier theorem on rank functions of connected matroids [15]; and Stastny recently showed that the tautological classes of Berget–Eur–Spink–Tseng [2], which interpolate between these two families, are extremal exactly for matroids that are connected up to loops and coloops [20].

In his thesis Huh introduced a second interpolating family, the *truncated Bergman fans* $\Delta_M[r_1, r_2]$ for $1 \leq r_1 \leq r_2 \leq \operatorname{rk} M - 1$ [11, Def. 48]. The weight of $\Delta_M[r_1, r_2]$ on the cone of a flag of flats $F_{r_1} \subsetneq \cdots \subsetneq F_{r_2}$, with $\operatorname{rk} F_i = i$, is $|\mu_M(\emptyset, F_{r_1})|$, and it is zero on all other cones. Huh proved that these weights are balanced [11, Prop. 49], so they define nef classes of dimension $r_2 - r_1 + 1$. He also observed that "the homology class of the truncated Bergman fan of a

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matroid often, but not always, generates an extremal ray of the nef cone". Examples 51 and 52 there are extremal, and Example 53 is a rank-4 matroid on six elements for which $\Delta_M[2,3]$ is a sum of two other extremal classes. Huh also gave a sufficient condition for extremality of two-dimensional classes in terms of a graph associated with the support [11, Prop. 41] and remarked that it is not necessary. The purpose of this paper is to understand which truncated Bergman fans are extremal.

Main results. Throughout, M is a loopless matroid of rank r on E, $\mathcal{L} = \mathcal{L}(M)$ is its lattice of flats, $\mathcal{L}_i$ is the set of flats of rank i, $W_i = |\mathcal{L}_i|$ are the Whitney numbers of the second kind, and $\mu(F) = \mu_M(\emptyset, F)$. We write $G \lessdot H$ when H covers G in $\mathcal{L}$.

Theorem A (One-dimensional truncations)**.** *Let $1 \le j \le r-1$. The truncated Bergman fan $\Delta_M[j,j]$ spans an extremal ray of $\mathrm{Nef}_1(X_E)$ if and only if either $j = 1$, or $j = r-1$ and $\mathcal{L}(M)$ is a modular lattice. More precisely, the space of one-dimensional Minkowski weights supported on the rays $\{e_F : F \in \mathcal{L}_j\}$ has dimension $W_j - W_1 + 1$.*

For $r_2 > r_1$ the picture is different: for instance $\Delta_M[2,3]$ is extremal for the uniform matroid $U_{4,5}$ although $\Delta_M[2,2]$ is not. The key structural fact is that the top rank plays no role.

Theorem B (Reduction)**.** *Let $1 \le r_1 < r_2 \le r-1$. Let $\Phi_{r_1}(M)$ be the space of functions $\varphi\colon \mathcal{L}_{r_1} \to \mathbb{R}$ such that for every $H \in \mathcal{L}_{r_1+1}$ there is a constant c_H with*

$$\sum_{\substack{G \in \mathcal{L}_{r_1} \\ P \le G \lessdot H}} \varphi(G) = c_H \qquad \text{for every } P \in \mathcal{L}_1 \text{ with } P \le H. \tag{1}$$

Then $\varphi \mapsto \big(F_{r_1} \subsetneq \cdots \subsetneq F_{r_2} \;\mapsto\; \varphi(F_{r_1})\big)$ is a linear isomorphism from $\Phi_{r_1}(M)$ onto the space of Minkowski weights supported on $\operatorname{supp}\Delta_M[r_1,r_2]$. In particular, $\Delta_M[r_1,r_2]$ is extremal if and only if $\dim \Phi_{r_1}(M) = 1$, and this condition does not depend on r_2.

The system (1) has $W_{r_1} + W_{r_1+1}$ unknowns, so extremality is decidable in polynomial time in the size of $\mathcal{L}$. Two consequences can be read off directly.

Theorem C (Sufficient conditions)**.** *Let $1 \le r_1 < r_2 \le r-1$.*

1. *$\Delta_M[1,r_2]$ is extremal. (It is the Bergman fan of the truncation of M to rank r_2+1.)*
2. *If $r_1 \ge 2$ and for every $H \in \mathcal{L}_{r_1+1}$ the restriction $M|H$ has a modular lattice of flats, then $\Delta_M[r_1,r_2]$ is extremal.*

Theorem D (Non-extremality)**.** *Let $1 \le r_1 < r_2 \le r-1$, and suppose that $\mathcal{S} \subseteq \mathcal{L}_{r_1}$ is a family such that for every $H \in \mathcal{L}_{r_1+1}$ each point of H lies in the same number of members $G \in \mathcal{S}$ with $G \lessdot H$. If the indicator function of $\mathcal{S}$ is not proportional to $|\mu|$ on $\mathcal{L}_{r_1}$, then $\Delta_M[r_1,r_2]$ is a sum of two non-proportional nonzero nef classes; in particular it is not extremal. For the graphic matroid of $K_{2,m}$, $m \ge 3$, the series pairs form such a family, so $\Delta_{M(K_{2,m})}[2,r_2]$ is not extremal for every $3 \le r_2 \le m$.*

Since $\operatorname{rk} M(K_{2,m}) = m+1$, Theorem D gives non-extremal examples with $r_1 = 2$ and $\operatorname{rk} M - r_1$ arbitrarily large.

Finally, we determined all extremal truncated Bergman fans of loopless matroids on at most seven elements, which is 1504 classes in total (Section 6). All of them agree with Theorems A–D. There are exactly five matroids with a non-extremal $\Delta_M[r_1,r_2]$ with $r_2 > r_1 \ge 2$. The smallest is $M(K_{2,3})$; it is the only one on six elements and therefore coincides with Huh's Example 53. The other four are obtained from $M(K_{2,3})$ by adding a parallel element, a coloop, a series element, or a non-graphic single-element extension.

Organization. Section 2 collects preliminaries on Minkowski weights and truncated Bergman fans. Section 3 proves Theorem A, Section 4 proves Theorems B and C, and Section 5 proves Theorem D. Section 6 describes the computations and Section 7 puts the results in the context of the full nef cone. Open questions are collected in Section 8.

2. Preliminaries

2.1. **The permutohedral fan and Minkowski weights.** Let $N_{\mathbb{R}} = \mathbb{R}^E/\mathbb{R}\mathbf{1}$, where $\mathbf{1} = \sum_{i\in E} e_i$, and for $S \subseteq E$ let e_S be the image of $\mathbf{1}_S = \sum_{i\in S} e_i$. For a chain $\mathcal{S} = (S_1 \subsetneq \cdots \subsetneq S_k)$ of nonempty proper subsets of E, let $\sigma_{\mathcal{S}}$ be the cone spanned by $e_{S_1}, \ldots, e_{S_k}$. These cones form the permutohedral fan Σ_E, which is complete and unimodular.

A *k-dimensional Minkowski weight* is a function w on chains of length k such that for every chain $\mathcal{T}$ of length $k-1$

$$\sum_S w(\mathcal{T} \cup \{S\})\, e_S \;\in\; \operatorname{span}\{e_T : T \in \mathcal{T}\}, \tag{2}$$

where the sum is over the subsets S such that $\mathcal{T} \cup \{S\}$ is a chain of length k. Because Σ_E is unimodular, e_S is a valid choice of lattice normal vector, so this is the balancing condition of [6]. Lifting to $\mathbb{R}^E$, (2) says that $\sum_S w(\mathcal{T} \cup \{S\})\mathbf{1}_S$ lies in the span of $\mathbf{1}_T$ ($T \in \mathcal{T}$) and $\mathbf{1}_E$. We will use repeatedly the following elementary fact.

Lemma 2.1. *Let $\emptyset = C_0 \subsetneq C_1 \subsetneq \cdots \subsetneq C_m = E$. A function $f\colon E \to \mathbb{R}$ lies in $\operatorname{span}\{\mathbf{1}_{C_1}, \ldots, \mathbf{1}_{C_m}\}$ if and only if f is constant on each layer $C_t \setminus C_{t-1}$.*

By [6], k-dimensional Minkowski weights are the elements of $A^{n-k}(X_E)_{\mathbb{R}}$, and $w(\sigma)$ is the degree of the product of the class with the orbit closure $V(\sigma)$. We write $\operatorname{Nef}_k(X_E)$ for the cone of nonnegative k-dimensional Minkowski weights; these are the classes of codimension $n-k$ that meet every torus-invariant subvariety of complementary dimension nonnegatively, i.e. the nef classes [11, 20].

Lemma 2.2 (cf. [20, Prop. 2.3])**.** *Let $w \in \operatorname{Nef}_k(X_E)$ be nonzero. Then w spans an extremal ray of $\operatorname{Nef}_k(X_E)$ if and only if every k-dimensional Minkowski weight v with $\operatorname{supp} v \subseteq \operatorname{supp} w$ is a scalar multiple of w.*

Proof. If v is supported in $\operatorname{supp} w$ and not proportional to w, then $w \pm \varepsilon v$ is nonnegative for small $\varepsilon > 0$, because w is positive on its support, and $w = \frac{1}{2}(w+\varepsilon v) + \frac{1}{2}(w-\varepsilon v)$ is a non-trivial decomposition. Conversely, if $w = w' + w''$ with w', w'' nef, then $\operatorname{supp} w' \subseteq \operatorname{supp} w$, so w' is a multiple of w. □

2.2. **Matroids.** We use [16] for matroid terminology. Since M is loopless, its points (rank-one flats) partition E, and for a flat F and $e \in E$ we have $e \in F$ if and only if $\operatorname{cl}(e) \le F$. The lattice $\mathcal{L}$ is geometric, so it is semimodular, and $\mu(F) = \mu_M(\emptyset, F)$ is nonzero with sign $(-1)^{\operatorname{rk} F}$ by Rota's theorem [19]. A lattice is *modular* if $\operatorname{rk} F + \operatorname{rk} G = \operatorname{rk}(F \vee G) + \operatorname{rk}(F \wedge G)$ for all F, G. For $s \le r$, the *truncation* $T^s(M)$ is the matroid of rank s whose proper flats are the flats of M of rank $< s$; its hyperplanes are the elements of $\mathcal{L}_{s-1}(M)$ and its points are those of M.

We use three classical results on Whitney numbers. Let I_j be the $W_j \times W_1$ incidence matrix of rank-j flats versus points.

Theorem 2.3 (Basterfield–Kelly [3], Greene [10])**.** *If $\mathcal{L}$ is a finite geometric lattice of rank $r \ge 2$, then $W_{r-1} \ge W_1$, with equality if and only if $\mathcal{L}$ is modular.*

Theorem 2.4 (Kung [13])**.** *If $\mathcal{L}$ is a finite geometric lattice of rank $r \ge 2$, the incidence matrix of hyperplanes versus points has rank W_1.*

Applying Theorem 2.4 to the truncation $T^{j+1}(M)$ shows that $\operatorname{rk} I_j = W_1$ for all $1 \le j \le r-1$.

2.3. **Truncated Bergman fans.**

Definition 2.5 ([11, Def. 48])**.** For $1 \le r_1 \le r_2 \le r-1$, the truncated Bergman fan $\Delta_M[r_1, r_2]$ is the function on chains of length $k = r_2 - r_1 + 1$ given by

$$\Delta_M[r_1, r_2](\mathcal{S}) = \begin{cases} |\mu(F_{r_1})| & \text{if } \mathcal{S} = (F_{r_1} \subsetneq \cdots \subsetneq F_{r_2}) \text{ with } F_i \in \mathcal{L}_i, \\ 0 & \text{otherwise.} \end{cases}$$

By [11, Prop. 49], $\Delta_M[r_1, r_2]$ is a k-dimensional Minkowski weight. Its support consists of the cones of flags of flats of consecutive ranks $r_1, \dots, r_2$. Note that $\Delta_M[1, r-1] = \Delta_M$ is the Bergman fan and, more generally, $\Delta_M[1, r_2]$ is the Bergman fan of $T^{r_2+1}(M)$, since $|\mu(P)| = 1$ for every point P.

3. One-dimensional truncations

Lemma 3.1. *Let $1 \le j \le r-1$. The space V_j of one-dimensional Minkowski weights supported on $\{e_F : F \in \mathcal{L}_j\}$ is isomorphic to $\{w \in \mathbb{R}^{\mathcal{L}_j} : I_j^{\mathsf{T}} w \in \mathbb{R}\mathbf{1}\}$, and $\dim V_j = W_j - W_1 + 1$.*

Proof. For $k = 1$, condition (2) is imposed only by the empty chain and says $\sum_F w_F \mathbf{1}_F \in \mathbb{R}\mathbf{1}_E$. Since the points partition E, $\mathbf{1}_F = \sum_{P \le F} \mathbf{1}_P$, and the vectors $\mathbf{1}_P$ are linearly independent with sum $\mathbf{1}_E$. Hence the condition is $\sum_{F \ge P} w_F = c$ for all points P and some constant c, i.e. $I_j^{\mathsf{T}} w = c\mathbf{1}$. The pair (w, c) is determined by w, so $\dim V_j = W_j + 1 - \operatorname{rk}[\, I_j^{\mathsf{T}} \mid \mathbf{1}\,]$. By Theorem 2.4 applied to $T^{j+1}(M)$, I_j^{T} has rank W_1, which is its number of rows, so $\operatorname{rk}[\, I_j^{\mathsf{T}} \mid \mathbf{1}\,] = W_1$. □

Proof of Theorem A. By Lemmas 2.2 and 3.1, $\Delta_M[j, j]$ is extremal if and only if $W_j = W_1$. This holds trivially for $j = 1$. For $j = r-1$ it holds if and only if $\mathcal{L}$ is modular, by Theorem 2.3.

Let $2 \le j \le r-2$, and let $T = T^{j+1}(M)$, a matroid of rank $j+1 \ge 3$ whose hyperplanes are the elements of $\mathcal{L}_j$. By Theorem 2.3, $W_j \ge W_1$, with equality only if $\mathcal{L}(T)$ is modular. We show that it is not. Since $r \ge j+2$, there are independent elements $x_1, \dots, x_{j+2}$ of M. Put $F = \operatorname{cl}(x_1, \dots, x_j)$ and $G = \operatorname{cl}(x_3, \dots, x_{j+2})$. Both lie in $\mathcal{L}_j$, so they are hyperplanes of T. Since $F \cup G$ has rank $j+2$ in M, their join in $\mathcal{L}(T)$ is E, of rank $j+1$. Submodularity in M gives $\operatorname{rk}(F \cap G) \le \operatorname{rk} F + \operatorname{rk} G - \operatorname{rk}(F \cup G) = j-2$. Modularity of $\mathcal{L}(T)$ would require $\operatorname{rk}(F \cap G) = 2j - (j+1) = j-1$. Hence $W_j > W_1$, and $\Delta_M[j, j]$ is not extremal. □

Remark 3.2. Modular geometric lattices are classical: the simplification of M has a modular lattice of flats if and only if it is a direct sum of points, lines, and projective geometries of rank at least 3 [4, Ch. IV]. Thus $\Delta_M[r-1, r-1]$ is extremal for Boolean matroids, projective geometries and their direct sums, and is not extremal for $U_{3,4}$, where the four points are covered by six lines. The strict inequality $W_j > W_1$ for $2 \le j \le r-2$ also follows from the Dowling–Wilson inequalities [5]; the argument above gives it directly.

4. The reduction theorem

Lemma 4.1. *Let $x \le y$ in a finite geometric lattice. Any two maximal chains of the interval $[x, y]$ can be joined by a sequence of maximal chains, each obtained from the previous one by replacing a single element.*

Proof. Induct on $\ell = \operatorname{rk} y - \operatorname{rk} x$; the cases $\ell \le 2$ are clear. Let $x \lessdot a_1 \lessdot \cdots$ and $x \lessdot b_1 \lessdot \cdots$ be maximal chains. If $a_1 = b_1$, apply induction to $[a_1, y]$. Otherwise, by semimodularity $c = a_1 \vee b_1$ covers a_1 and b_1. Choose a maximal chain C of $[c, y]$. The chains (x, a_1, c, C) and (x, b_1, c, C) differ in one element. The first is connected to $(x, a_1, \dots)$ by induction in $[a_1, y]$, and the second to $(x, b_1, \dots)$ by induction in $[b_1, y]$. □

Proof of Theorem B. Let $k = r_2 - r_1 + 1 \geq 2$, let $Z = \operatorname{supp} \Delta_M[r_1, r_2]$, and let v be a k-dimensional Minkowski weight with $\operatorname{supp} v \subseteq Z$. Every codimension-one face of a cone in Z is obtained by omitting one flat F_i from a flag $F_{r_1} \subsetneq \cdots \subsetneq F_{r_2}$. If $\mathcal{T}$ is such a face, the cones of Z containing it are obtained by inserting a flat of rank i into the gap; balancing at all other codimension-one cones is automatic because v vanishes near them.

Step 1: omitting a flat above the bottom. Let $r_1 < i \leq r_2$ and let $\mathcal{T}$ be the flag with F_i omitted. Set $U = F_{i+1}$ if $i < r_2$ and $U = E$ if $i = r_2$. The flats $G \in \mathcal{L}_i$ that can be inserted are those with $F_{i-1} \lessdot G \subseteq U$. The sets $G \setminus F_{i-1}$ partition $U \setminus F_{i-1}$, because each $e \in U \setminus F_{i-1}$ lies in exactly one of them, namely $\operatorname{cl}(F_{i-1} \cup e)$. Since $\mathbf{1}_G = \mathbf{1}_{F_{i-1}} + \mathbf{1}_{G \setminus F_{i-1}}$, condition (2) at $\mathcal{T}$ says that $\sum_G v(\mathcal{T} \cup \{G\}) \mathbf{1}_{G \setminus F_{i-1}}$ lies in the span of the indicators of the chain $\mathcal{T} \cup \{E\}$. In that chain, F_{i-1} and U are consecutive, so by Lemma 2.1 the sum must be constant on $U \setminus F_{i-1}$. Since the blocks are nonempty and partition $U \setminus F_{i-1}$, this holds exactly when all the values $v(\mathcal{T} \cup \{G\})$ are equal.

Thus v is unchanged when one flat F_i with $i > r_1$ is replaced by another with the same neighbours (for $i = r_2$, with the same F_{r_2-1}). Fix F_{r_1} and extend two flags with bottom F_{r_1} to maximal chains of $[F_{r_1}, E]$. By Lemma 4.1 they are joined by single-element replacements. A replacement at rank $> r_2$ does not change the truncated flag, and a replacement at rank $i \in (r_1, r_2]$ is one of the moves above. Hence $v(F_{r_1} \subsetneq \cdots \subsetneq F_{r_2}) = \varphi(F_{r_1})$ for a function φ on $\mathcal{L}_{r_1}$. Every $G \in \mathcal{L}_{r_1}$ is the bottom of a flag in Z, so φ is determined by v.

Step 2: omitting the bottom flat. Let $\mathcal{T} = (F_{r_1+1} \subsetneq \cdots \subsetneq F_{r_2})$ and $H = F_{r_1+1}$. The insertable flats are the $G \in \mathcal{L}_{r_1}$ with $G \lessdot H$, and (2) reads $\sum_{G \lessdot H} \varphi(G) \mathbf{1}_G \in \operatorname{span}\{\mathbf{1}_H, \mathbf{1}_{F_{r_1+2}}, \ldots, \mathbf{1}_{F_{r_2}}, \mathbf{1}_E\}$. The left side vanishes outside H, and H is the first layer of the chain, so by Lemma 2.1 this is equivalent to $\sum_{G \lessdot H} \varphi(G) \mathbf{1}_G = c_H \mathbf{1}_H$. Evaluating at $e \in H$ and using $e \in G \iff \operatorname{cl}(e) \leq G$ gives exactly (1) for the point $P = \operatorname{cl}(e)$. The condition depends only on H, and every $H \in \mathcal{L}_{r_1+1}$ occurs as the bottom of such a $\mathcal{T}$ because $r_1 + 1 \leq r_2 \leq r - 1$.

Conversely, if $\varphi \in \Phi_{r_1}(M)$, the weight $v(F_{r_1} \subsetneq \ldots) = \varphi(F_{r_1})$ satisfies the conditions of both steps, hence is a Minkowski weight. This gives the isomorphism. Since $\Delta_M[r_1, r_2]$ corresponds to $\varphi = |\mu|$, the last statement follows from Lemma 2.2. □

Remark 4.2. Theorem B together with [11, Prop. 49] shows that $|\mu|$ always satisfies (1). Comparing with Lemma 3.1, (1) says that for each $H \in \mathcal{L}_{r_1+1}$ the restriction of φ to the hyperplanes of $M|H$ is a one-dimensional Minkowski weight of $M|H$ in the sense of Section 3. Thus Theorem B reduces extremality to a gluing problem for the local spaces $V_{r_1}(M|H)$, $H \in \mathcal{L}_{r_1+1}$, which have dimension $W_{r_1}(M|H) - W_1(M|H) + 1$ by Lemma 3.1.

Proof of Theorem C. (1) Here $\mathcal{L}_{r_1}$ is the set of points and the only point G with $P \leq G \lessdot H$ is P itself, so (1) says that φ is constant on the points of each $H \in \mathcal{L}_2$. Any two points lie in a common rank-two flat, so φ is constant and $\dim \Phi_1(M) = 1$.

(2) Fix $H \in \mathcal{L}_{r_1+1}$. The restriction of any $\varphi \in \Phi_{r_1}(M)$ to $\{G \in \mathcal{L}_{r_1} : G \lessdot H\}$ lies in the local space of the remark above. By Lemma 3.1 and Theorem 2.3 applied to $M|H$, whose lattice of flats $[\emptyset, H]$ is modular of rank $r_1 + 1 \geq 3$, this space has dimension one. Since $|\mu|$ belongs to it, the restriction is $\lambda_H |\mu|$ for some $\lambda_H \in \mathbb{R}$. If $H, H' \in \mathcal{L}_{r_1+1}$ cover a common $G \in \mathcal{L}_{r_1}$, then $\lambda_H |\mu(G)| = \varphi(G) = \lambda_{H'} |\mu(G)|$, and $\mu(G) \neq 0$ gives $\lambda_H = \lambda_{H'}$. The graph on $\mathcal{L}_{r_1+1}$ with these adjacencies is connected. Indeed, take maximal chains of $\mathcal{L}$ through H and H' and join them as in Lemma 4.1. Each replacement either fixes the element of rank $r_1 + 1$ or replaces it while fixing the element of rank r_1, which is then a common cover. Hence all λ_H are equal. Every $G \in \mathcal{L}_{r_1}$ is covered by some H, so $\varphi = \lambda |\mu|$. □

Remark 4.3. For $r_1 \geq 2$ the cases $r_2 = r_1$ and $r_2 > r_1$ behave very differently. By Theorem A, $\Delta_M[r_1, r_1]$ is never extremal when $2 \leq r_1 \leq r-2$, while by Theorem C(2) $\Delta_M[r_1, r_2]$ is extremal

for every $r_2 > r_1$ when, for example, all rank-(r_1+1) flats of M are Boolean. This happens for every uniform matroid $U_{r,m}$ with $r_1 \leq r-2$.

5. Non-extremality

Proof of Theorem D. The hypothesis on $\mathcal{S}$ says exactly that its indicator function $\mathbf{1}_{\mathcal{S}}$ satisfies (1). Let $w_{\mathcal{S}}$ be the corresponding Minkowski weight from Theorem B: it equals 1 on flags whose bottom lies in $\mathcal{S}$ and 0 elsewhere. It is nef. Let $t = \min_{G\in\mathcal{S}} |\mu(G)|$; the family $\mathcal{S}$ is nonempty because $\mathbf{1}_{\mathcal{S}}$ is not proportional to $|\mu|$. Then $\Delta_M[r_1,r_2] - t\,w_{\mathcal{S}}$ is a nonnegative Minkowski weight. It is nonzero and not proportional to $w_{\mathcal{S}}$, since otherwise $|\mu|$ would be proportional to $\mathbf{1}_{\mathcal{S}}$. Hence $\Delta_M[r_1,r_2] = t\,w_{\mathcal{S}} + (\Delta_M[r_1,r_2] - t\,w_{\mathcal{S}})$ is a sum of two non-proportional nonzero nef classes.

Now let $M = M(K_{2,m})$ with $m \geq 3$. Let a, b be the vertices of degree m and $u_1, \ldots, u_m$ those of degree two. Then $\operatorname{rk} M = m+1$, every pair of edges is a flat of rank two (the graph is simple and has no triangles), and $|\mu(G)| = 1$ for all $G \in \mathcal{L}_2$. Let $\mathcal{S} = \{\{au_i, bu_i\} : 1 \leq i \leq m\}$ be the set of series pairs. This is a proper nonempty subset of $\mathcal{L}_2$, so $\mathbf{1}_{\mathcal{S}}$ is not proportional to $|\mu|$.

Let $H \in \mathcal{L}_3$. If H contains no series pair, both sides of the required condition are 0. If $\{au_i, bu_i\} \subseteq H$, choose an edge $f \in H$ outside this pair. Then $f \in \{au_j, bu_j\}$ for some $j \neq i$, so H contains the 4-cycle $C = \{au_i, bu_i, au_j, bu_j\}$, which has rank 3, and therefore $H = \operatorname{cl}(C)$. Adding any further edge au_l or bu_l with $l \neq i, j$ to C raises the rank, so C is a flat and $H = C$. The members of $\mathcal{S}$ contained in C are the two series pairs $\{au_i, bu_i\}$ and $\{au_j, bu_j\}$, which partition C. Hence every point of H lies in exactly one member of $\mathcal{S}$ below H. The first part of the theorem applies for every $3 \leq r_2 \leq m = \operatorname{rk} M - 1$. □

Example 5.1 (Huh's Example 53)**.** For $M = M(K_{2,3})$ (rank 4 on six elements), the decomposition in the proof is $\Delta_M[2,3] = w_{\mathcal{S}} + (\Delta_M[2,3] - w_{\mathcal{S}})$. The first summand is supported on the 6 flags whose bottom is a series pair and the second on the remaining 36 flags. By the computations of Section 6, $M(K_{2,3})$ is the only loopless matroid on at most six elements with a non-extremal $\Delta_M[r_1,r_2]$, $r_2 > r_1 \geq 2$, and a direct computation with Lemma 2.2 shows that both summands above are extremal. This matches the description of [11, Ex. 53]: a rank-4 matroid on $\{0,\ldots,5\}$ whose $\Delta_M[2,3]$ is a non-trivial sum of two extremal nef classes. Therefore Huh's example is $M(K_{2,3})$, up to isomorphism (matroids in [11] are loopless).

6. Computations

We enumerated all loopless matroids of rank at least two on at most seven elements up to isomorphism. Matroids of rank at most three were constructed from a partition into parallel classes together with a linear space of lines, matroids of rank at least $|E|-3$ by duality, and the free matroid was added. The total numbers of isomorphism classes of all matroids on $1,\ldots,7$ elements agree with the known values $2, 4, 8, 17, 38, 98, 306$ [14]. For every matroid and every $1 \leq r_1 \leq r_2 \leq r-1$ we computed the flats, checked that $\Delta_M[r_1,r_2]$ is balanced, and computed the dimension of the space of Minkowski weights supported on its support exactly over $\mathbb{Z}$ using FLINT [9]. By Lemma 2.2, extremality is equivalent to this dimension being one. Independently, we computed $\dim \Phi_{r_1}(M)$ from (1). The two computations agreed in all 739 cases with $r_2 > r_1$, and in all 765 cases with $r_2 = r_1$ the dimension equalled $W_j - W_1 + 1$.

All balancing checks succeeded. The five non-extremal cases with $r_2 > r_1 \geq 2$ are:

(i) $M(K_{2,3})$, with $\Delta_M[2,3]$;
(ii) $M(K_{2,3})$ with one edge doubled (rank 4, seven elements), with $\Delta_M[2,3]$;
(iii) $M(K_{2,3}) \oplus U_{1,1}$, the graphic matroid of $K_{2,3}$ plus a pendant edge (rank 5), with $\Delta_M[3,4]$;

$\lvert E\rvert$	matroids	$\Delta_M[1,1]$	$\Delta_M[j,j]$, $j \geq 2$	$\Delta_M[1,r_2]$, $r_2 \geq 2$	$\Delta_M[r_1,r_2]$, $r_2 > r_1 \geq 2$
3	3	3/3	1/1	1/1	–
4	8	8/8	3/5	5/5	1/1
5	20	20/20	8/20	20/20	7/7
6	59	59/59	19/80	80/80	38/39
7	207	207/207	41/362	362/362	220/224
total	297	297/297	72/468	468/468	266/271

TABLE 1. Extremal/total numbers of truncated Bergman fans of loopless matroids of rank ≥ 2 on $\lvert E\rvert \leq 7$ elements, up to isomorphism. The column $j \geq 2$ consists exactly of the cases $j = r-1$ with $\mathcal{L}(M)$ modular, in accordance with Theorem A.

(iv) the graphic matroid of the theta graph with paths of lengths $2, 2, 3$, a series extension of $M(K_{2,3})$ (rank 5), with $\Delta_M[3,4]$;
(v) a non-graphic single-element extension of $M(K_{2,3})$ of rank 4 with exactly one 3-circuit, such that deleting any element of the 3-circuit gives $M(K_{2,3})$, with $\Delta_M[2,3]$.

In each case $\dim \Phi_{r_1}(M) = 2$. Every example contains $M(K_{2,3})$ as a minor, and in this range every failure has $r_1 = r-2$. Theorem D shows that the latter is an artifact of small ground sets: by direct computation $\dim \Phi_2(M(K_{2,m})) = 2$ for $m = 3, 4, 5$, and for $M(K_{2,4})$ (eight elements, rank 5) the weights $\Delta_M[2,3]$ and $\Delta_M[2,4]$ are non-extremal with $r_1 = r-3$. In cases (i), (ii), (iii) and (v) the second solution of (1) can be taken to be the indicator function of a family $\mathcal{S}$ as in Theorem D; we did not determine whether this is possible in case (iv).

7. Context: the full nef cone

Truncated and ordinary Bergman fans form a small part of the extremal rays of $\mathrm{Nef}_k(X_E)$. We record two computations for $k = 2$.

For $n = 3$, two-dimensional Minkowski weights on Σ_E are divisor classes, and $\mathrm{Nef}_2(X_{A_3})$ is the nef cone of divisors of the threefold X_{A_3}. It is isomorphic to the cone of submodular functions on $2^{\{0,1,2,3\}}$ modulo modular functions (see e.g. [17]), which has 37 extremal rays by Promislow and Young [18]. Our exact computation (Normaliz [8] and cdd [7]) confirms 37. All 37 take only the values 0 and 1. Eleven are Bergman fans of loopless rank-three matroids on four elements, five are images of Bergman fans under the Cremona involution $S \mapsto E \setminus S$, and 21 are neither.

For $n = 4$, $\mathrm{Nef}_2(X_{A_4})$ is a cone of dimension 66 in $\mathbb{R}^{150}$; it parametrizes classes of codimension two on a fourfold. A complete enumeration was out of reach of Normaliz and cdd on our hardware. Sampling vertices of random linear programs over a compact slice, and certifying each one exactly via Lemma 2.2, produced 939,807 distinct extremal rays in 4,439 orbits under $S_5 \times \mathbb{Z}/2$, and the number found was still growing when we stopped. Only 9 of these orbits contain Bergman fans of matroids. The others have weights up to 16 and supports of 6 to 60 cones. So $\mathrm{Nef}_2(X_{A_4})$ has at least 939,807 extremal rays, and almost all are non-matroidal.

8. Questions

Question 8.1. Give a combinatorial characterization of the matroids M and ranks $r_1 \geq 2$ with $\dim \Phi_{r_1}(M) = 1$.

Question 8.2. If $\Delta_M[r_1, r_2]$ is not extremal, does $\Phi_{r_1}(M)$ always contain the indicator function of a family $\mathcal{S}$ as in Theorem D? In other words, is the second solution of (1) always "combinatorial"?

Question 8.3. In all examples on at most seven elements, non-extremality with $r_2 > r_1 \geq 2$ comes with an $M(K_{2,3})$-minor. Is there a meaningful relation between non-extremality and minors? Is $\dim \Phi_{r_1}(M)$ bounded, for instance by the number of connected components of a suitable auxiliary structure?

Question 8.4. Determine the number of extremal rays of $\mathrm{Nef}_2(X_{A_4})$.

Code and data. The code used for all computations, together with the resulting data, is available at [repository URL].

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